

Advanced Mobile Communication Networks

Course Description and Introduction

SS 2024

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Ilmenau University of Technology

Outline

- Motivation
- Objectives
- Organizational Stuff
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Motivation

Why is it important?

- The Internet and mobile communication networks are experiencing an enormous growth and are a crucial factor of today's economy

Today's networks

- Heterogeneous fixed and mobile networks interconnected by a common IP core
- Quick introduction of new network elements, applications and services (software-defined networking, software-defined radio)
- Internet of Things (V2V/V2X, M2M, D2D)

Research issues:

- Higher bandwidth, reduced delay, higher reliability, lower power consumption, lower cost, security

Challenges:

- Heterogeneity, dynamics, complexity, cloudification, scalability, QoS
- Energy, security, network management

Objectives

- Provide an in-depth understanding of research issues in mobile communication networks
- Understand protocol and networking issues of future mobile communication networks, esp. ad hoc networks and cognitive radio systems
- Get prepared for research in mobile networking

This is a Master course!

- Material is not given to simply memorize!
- Digesting the material is the most important part!
 - Why is it done that way? Why not differently?
 - What are the alternatives? What are the pros and cons?
 - ...

Organizational Stuff

- Course prerequisites
 - Required: basics of communication networks and protocols
 - Strongly recommended: basics of IP-based mobile communication systems, e.g. Mobilkommunikationsnetze (B.S. program)
- Course material is provided in Moodle
⇒ register yourself under moodle2.tu-ilmenau.de
- Course budget: 5 credits (roughly 2+2 SWS)
- Quizzes are recommended but do not account for the grade
- 10% additional bonus points can be earned by each of two bonus exams, i.e. 20% altogether
- Exam and grading scheme
 - Written final exam
 - Bonus points only count if a passing grade is achieved in the exam!
 - Bonus points are valid for the current and the following semester

Instructor Contact information

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Course Units for Offline Studies

- Introduction
- Wireless Transmission
- Medium Access Schemes
- IEEE 802.11 Networks
- Mobility Management
- Transport Layer
- Quality of Service (QoS)
- Self-Organization
- Ad Hoc Networks
- Cognitive Radio Networks
- Cellular Networks and Network Evolution



Class Schedule (provisional)

Index	Lecture	Quizzes	Time	Date	Lecturer	Room
1	Introduction		9:00-10:30	Apr. 10	Mts	HS 2
2.1	Wireless Transmission	1-4	11:00-12:30	Apr. 10	Mts	HS 2
2.2	Wireless Transmission	5-7	9:00-10:30	Apr. 17	Mts	HS 2
3	Medium Access schemes	all	11:00-12:30	Apr. 17	Mts	HS 2
	Seminar Top 2&3		9:00-12:30	Apr. 24	Ali, Faiaz	HS 2
4	IEEE 802.11	all	9:00-10:30	May 8	Mts	HS 2
5.1	Mobility Management I	1-3	11:00-12:30	May 8	Ali	HS 2
5.2	Mobility Management II	4, 5	9:00-10:30	May 15	Ali	HS 2
	Seminar Top 4		11:00-12:30	May 15	Faiaz	HS 2
	Seminar Top 5		9:00-10:30	May 22	Ali	HS 2
	Bonus Exam Top 2-5		11:00-12:30	May 22	Ali, Faiaz	HS 2
6.1	Transport Protocols I	1, 2	9:00-10:30	May 29	Mts	HS 2
6.2	Transport Protocols II	3, 4	11:00-12:30	May 29	Mts	HS 2
7.1	Quality of Service I	1-3	9:00-10:30	June 5	Mts/Ali	HS 2
7.2	Quality of Service II	4-7	11:00-12:30	June 5	Mts/Ali	HS 2
8	Self-organization	all	9:00-10:30	June 12	Mts/Ali	HS 2
	Seminar Top 6 & 7		11:00-12:30	June 12	Faiaz	HS 2
	Dies Academicus			June 19		
9	Ad Hoc Networks	1,2	9:00-10:30	June 26	Mts	HS 2
10	Cognitive Radio Networks	2	11:00-12:30	June 26	Mts	HS 2
	Seminar Top 8 & 9		9:00-10:30	July 3	Ali, Faiaz	HS 2
	Bonus Exam Top 6-9		11:00-12:30	July 3	Ali, Faiaz	HS 2
11.1	Cellular Networks I	1	9:00-10:30	July 10	Mts	HS 2
11.2	Cellular Networks II	2, 3	11:00-12:30	July 10	Mts	HS 2
	Seminar Top 10, 11 and Wrap Up		9:00-12:30	July 17	Mts, Ali, Faiaz	HS 2

Background: Overview on Courses Provided by ICS

B.Sc.

1.

Techn. Informatik 1/
Rechnerorganisation

4.

Integrierte Hard- und
Softwaresysteme

5.

Mobilkommunikationsnetze

Entwicklung integrierter
HW/SW-Systeme (IHS2)

6.

M.Sc.

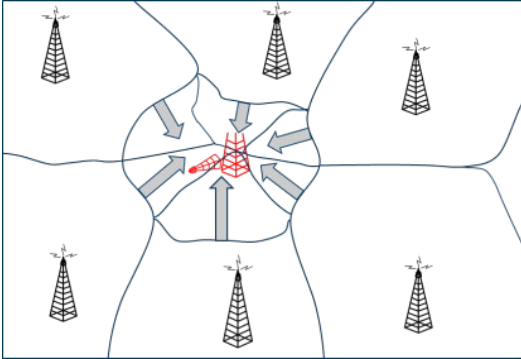
Advanced Mobile Communication Networks
(SS)
MSCSP, RCSE, II, IN

Spezielle Aspekte integr.
HW/SW-Systeme (SS)

Cellular Communication Systems (WS)
IN, II, MSCSP, RCSE

Doctoral Studies ...

Background: Core ICS Research Areas



5G and Beyond (6G Campus Ilmenau)

- Machine-learning applied network optimization
- Orchestration of private 5G networks
- Resource management and energy saving
- Mission-Critical Communication (MCC)
- Multipath communication
- Device-to-device communication
- Low-latency, high reliability communication



Communicating Unmanned Aerial Vehicles (UAVs)

- Autonomous, safe and collision-free operation
- Linux-based system for easy extendability
- Swarm operation
- Identification of coverage and service problems
- Provision of real-time and delay-tolerant communications

Outline of Questions for Entry Quiz

1. IP protocol model (similar to OSI protocol layers)
 - a. What is the purpose of layering or protocols?
 - b. Draw the layered architecture (IP or OSI)!
 - c. What's the difference between a protocol function and a service of a protocol?
 - d. Which protocol layers provide end-to-end, which hop-by-hop services?
2. What's the functionality provided by the (a) network layer, (b) transport layer, (c) link layer and (d) physical layer?
3. What's the difference between the transport protocol and the logical link sublayer?
4. What is the purpose of a TCP/UDP port and what is the difference to the IP address?
5. What does CSMA stand for? What's the idea and how does it work?
6. What's the purpose of IP subnetting? What's the effect on routing?
7. What's the difference between MAC and IP addresses?
8. What does VPN mean? Which protocol implements it?
9. What is the difference between VoIP and circuit-switched voice?

Online Entry Quiz

- For less than 80% correct answers:
 - take a course on communication networks, e.g. course by Prof. Dr. Jochen Seitz on Communication Networks (see CSP program)
 - self-study of communication networks (Coursera, udemy, etc.)

References on Background Info (for self-studies)

Communication networks and protocols

- Peterson & Davie: Computer Networks: A Systems Approach, 5th edition Morgan Kaufmann, 2012 (problem-oriented, entertaining bottom-up introduction in networking and communication protocols)
- A. Tannenbaum: Computer Networks, 5th edition, Pearson Education, 2012 (detailed bottom-up description of protocols & networking)
- K.R. Fall, W.R. Stevens: TCP/IP Illustrated, Vol. 1: The Protocols, 2nd edition, Addison-Wesley, 2011 (detailed reference for TCP/IP protocol family)
- J. Kurose, K. Ross: Computer Networking: A Top-down Approach, Pearson, 2013 (application-oriented introduction in networking)
- W. Stallings: Data and Computer Communications, 10th edition, Pearson Education, 2013 (comprehensive textbook on wired and wireless communications)
- Lots of good videos on youtube, just search for computer networking, network protocols, TCP/IP or OSI model, e.g.
<https://www.youtube.com/watch?v=svkGASq8mNM> or
https://www.youtube.com/watch?v=3b_TAYtzuho